LESSON BOOKLET • YEARS 7 AND 8

Forces and Motion

Types of force • Newton's laws • energy • levers • projectiles

Victorian Curriculum F–10 Version 2.0 | VC2S8U13 | VC2S8U14 | VC2S8U15 | VC2S8I01–I08

Name: _____ **Class:** _____

Teacher: _____ **Year level:** _____

How to use this booklet

There is one section for each lesson, and every lesson works the same way:

Part A — Reading	a short passage and comprehension questions to start the lesson
Part B — Theory	the ideas, with worked examples to follow step by step
Part C — Practice	questions like the examples, so you can build the skill
Part D — Above level	harder, higher-order questions — Year 8 and extension

Experiment lessons follow the same four parts: a reading about the science, planning the experiment, results and analysis, then above-level questions. Everything leads to the trebuchet investigation at the end of the unit.

Lesson	Title	Yr 7 marks	Yr 8 marks	Page
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2	Force Diagrams and Net Force	31	45	p. 9
3	Experiment 1 — Investigating Friction	13	20	p. 16

1 What is a Force?

Pushes and pulls • Measuring force • Mass and weight • Contact and non-contact forces

Victorian Curriculum 2.0: **VC2S8U14 • VC2S8I04**

Part A • Reading and comprehension

Reading — The Forces You Cannot See

Right now, while you sit perfectly still, you are being pushed and pulled from several directions at once. Gravity is pulling every part of your body downwards towards the centre of the Earth. At the same time, the chair beneath you is pushing upwards on you with exactly the same strength. Because these two forces are equal and opposite, you stay put. If the chair suddenly vanished, the pushing force would disappear, gravity would win, and you would fall.

People often think that astronauts on the International Space Station float because there is no gravity in space. This is not true. At the height the Space Station orbits, Earth's gravity is still about 90% as strong as it is on the ground. Astronauts float because they — and the whole Space Station — are falling around the Earth at enormous speed, and they fall together. Everything inside falls at the same rate, so nothing presses against anything else. The astronauts do not feel the push of a floor, so they seem weightless.

Forces are invisible, so scientists need a way to record them. The solution is simple but powerful: draw each force as an arrow. The direction of the arrow shows which way the force acts, and the length of the arrow shows how strong it is. A picture full of invisible pushes and pulls becomes something you can measure, compare and calculate with.

C1. According to the passage, name the two forces acting on you while you sit still, and state the direction of each.

C2. The passage says that if the chair vanished, "gravity would win". Explain what this means, using the idea of two forces that are no longer equal.

C3. Explain why astronauts on the Space Station appear to float, even though gravity is still acting on them.

C4. Does an astronaut on the Space Station still have weight? Use a number from the passage to support your answer.

C5. The passage says drawing forces as arrows is "simple but powerful". What two pieces of information does a force arrow give you?

Part B • Theory and examples

1.1 Every Force is a Push or a Pull

This is the whole idea, and it really is that simple. Whenever one object affects another — by touching it, or from a distance — it does so by **pushing** it or **pulling** it. There is no third option.

Think about it

A magnet *pulls* a paperclip. A hand *pushes* a door. The Earth *pulls* you downwards. A rope *pulls* a sledge. The ground *pushes* up on your feet.

1.2 What Forces Can Do

A force can only change an object in a small number of ways. Learn these five and you can describe almost any situation in this unit.

#	A force can…	Example
1	Make a still object start moving	A kicked football leaves the ground
2	Make a moving object stop	Brakes bring a bicycle to a halt
3	Make an object speed up or slow down	Pedalling harder makes you accelerate
4	Change the direction of motion	A tennis racquet sends the ball back
5	Change the shape of an object	Squashing a drink can

1.3 Measuring Force: The Newton

Force is measured in **newtons**, written **N**. The unit is named after Sir Isaac Newton, whose laws of motion you will meet in Lesson 4.

Getting a feel for a newton

1 N is roughly the force you feel holding a small apple (about 100 g) against gravity.
10 N is about the weight of a 1 kg bag of sugar.
500 N is roughly the weight of an average adult.

We measure force with a **spring scale** (also called a *newton meter*). Inside is a spring: the harder you pull, the further it stretches, and a pointer moves down a scale marked in newtons.

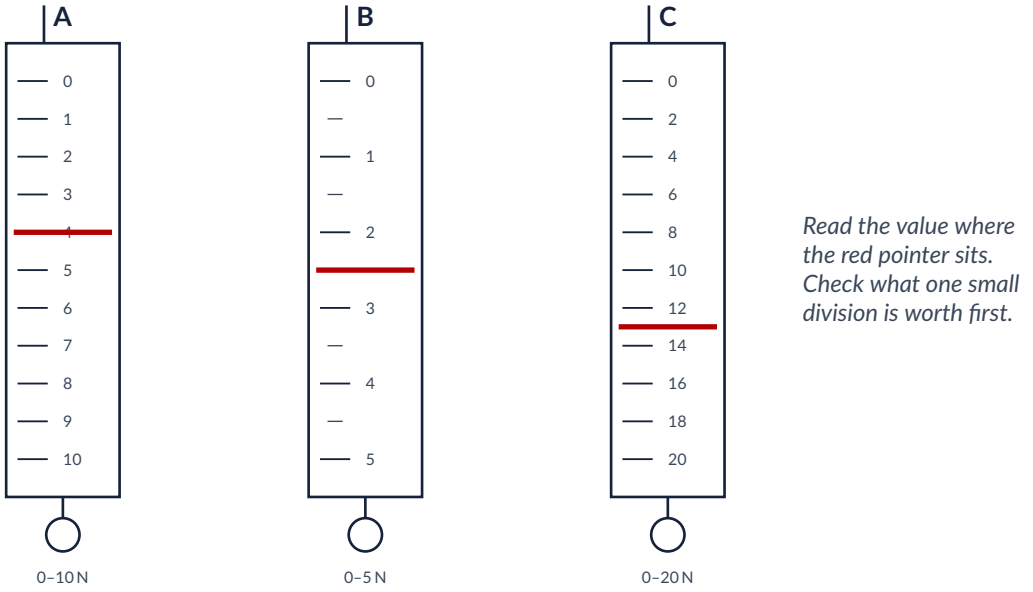


Figure 1.1 — Reading a spring scale. Always read at eye level, and check the size of each small division first.

How to read a spring scale correctly

1. Check the **largest** number on the scale — this is the maximum it can measure.
2. Work out what **one small division** is worth. If there are 10 small gaps between 0 and 5 N, each is 0.5 N.
3. Hold the scale still and read at **eye level** to avoid parallax error.
4. Read the value where the **pointer** sits, and always write the **unit (N)**.

Worked Example 1.1 — Reading a spring scale

A spring scale is marked from 0 to 10 N, with **20** small divisions between 0 and 10. The pointer sits on the **7th** small division. What is the reading?

Step 1 — one division: $10 \div 20 = 0.5$ N per division
Step 2 — the reading: $7 \times 0.5 = \mathbf{3.5}$ N

1.4 Mass and Weight Are Not the Same Thing

In everyday speech we use these words as if they mean the same thing. In science they are quite different, and mixing them up causes a lot of lost marks.

	Mass	Weight
What it is	The amount of matter in an object	The *force* of gravity pulling on that mass
Unit	kilograms (kg)	newtons (N)
Measured with	A balance	A spring scale
Changes?	Same everywhere in the universe	Changes with the strength of gravity

The link between them

$$\text{Weight} = \text{mass} \times \text{gravitational field strength} \quad \Rightarrow \quad \boxed{W = mg}$$

On Earth, $g = 9.8$ N/kg. On the Moon, $g = 1.6$ N/kg.

Worked Example 1.2 — Mass and weight

A student has a mass of 50 kg. Find their weight on Earth, and on the Moon.

On Earth: $W = mg = 50 \times 9.8 =$ **490** N
On the Moon: $W = mg = 50 \times 1.6 =$ **80** N

Their *mass* is 50 kg in both places — only the *weight* changes.

1.5 Contact and Non-contact Forces

Forces fall into two groups. **Contact forces** only act when two objects touch. **Non-contact forces** act across a gap — the objects do not need to touch at all.

Contact Forces
- Friction
- Normal force
- Tension
- Applied force
- Air resistance
- Spring force

Non-contact Forces
- Gravity
- Magnetic force
- Electrostatic force
- Nuclear force

Figure 1.2 — Contact and non-contact forces

Worked Example 1.3 — Classifying forces

A fridge magnet holds a shopping list on the fridge door without sliding down. Name two forces acting on the magnet and classify each one.

Weight pulls the magnet down — gravity acts across a gap, so it is a **non-contact** force.
Friction between the magnet and the door stops it sliding — the surfaces must touch, so it is a **contact** force.

(The magnetic pull towards the door is also non-contact — it is what presses the magnet on hard enough for friction to hold it.)

Part C • Practice

Work through every task. Show your working for calculations, and always include the unit.

Task 1.1 *(6 marks)* For each everyday action, decide whether the force is a push or a pull. Write **PUSH** or **PULL** in the box.

Action	Push or pull?	Action	Push or pull?
Kicking a football		Opening a drawer	
A crane lifting a girder		Squeezing a stress ball	
A dog straining on a lead		Wind on a sail	

Task 1.2*(4 marks)* For each event below, write the **number** (1–5) of the effect the force is having (see the table in §1.2). Some may have more than one.

A goalkeeper catches a ball	Stretching a rubber band
A car turns a corner at steady speed	A rocket lifts off

Task 1.3 *(3 marks)* Write the reading shown on each of the three scales in Figure 1.1.

Scale A	Scale B	Scale C

Task 1.4 *(4 marks)* Complete the table. Use $g = 9.8$ N/kg on Earth and $g = 1.6$ N/kg on the Moon.

Object	Mass (kg)	Weight on Earth (N)	Weight on Moon (N)
Bag of sugar	1		
Bicycle	12		
Small car	900		

Task 1.5 *(6 marks)* Complete the table. For each force, write whether it is a contact or non-contact force, and give a real-world example.

Force	Contact / Non-contact	Real-world example
Gravity		
Friction		
Magnetic force		
Normal force		
Air resistance		
Tension		

Task 1.6 *(2 marks)* A spring scale is marked from 0 to 20 N, with **40** small divisions. The pointer sits on the **13th** small division. What is the reading? Show both steps.

Task 1.7 *(2 marks)*
A cat has a mass of 4 kg. Find its weight on Earth.

Task 1.8 *(2 marks)*
The same cat travels to the Moon. Find its weight there. Has its mass changed?

Task 1.9 *(2 marks)* A balloon rubbed on your hair sticks to a wall. Name the force holding it there, and say whether it is a contact or non-contact force.

Part D • Above level

These questions go beyond the Year 7 standard. They ask you to combine ideas, use a formula, or judge someone else's reasoning.

Extension 1.10 *(4 marks)* An astronaut has a mass of 75 kg. **(a)** Find her weight on Earth. **(b)** Find her weight on the Moon. **(c)** Explain why she can jump much higher on the Moon, even though her mass has not changed.

Extension 1.11 *(3 marks)* A suitcase has a weight of 196 N on Earth. **(a)** Find its mass. **(b)** Find its weight on the Moon.

Extension 1.12 *(3 marks)* A bathroom scale really measures *force*, but it is labelled in kilograms. A 60 kg student reads 60 kg on it on Earth. Predict what the same scale would read on the Moon, and explain why the scale is not really measuring mass.

2 Force Diagrams and Net Force

Naming forces • Force arrows • Free body diagrams • Balanced and unbalanced forces

Victorian Curriculum 2.0: **VC2S8U14 • VC2S8I04**

Part A • Reading and comprehension

Reading — Winning a Tug-of-War

In a tug-of-war, two teams pull on the same rope in opposite directions. While the rope stays still, the two pulls are exactly the same size. They are *balanced*, so the net force on the rope is zero — the rope does not move, even though both teams are pulling as hard as they can.

The moment one team pulls even a little harder, the forces become *unbalanced*. There is now a net force towards the stronger team, and the rope starts to move their way. The bigger the difference between the two pulls, the faster the rope speeds up.

So how does a team pull harder? Strength helps, but the real secret is under their feet. To pull on the rope, each person pushes backwards against the ground with their shoes. Friction between the shoes and the ground stops them from sliding forwards. A team with good grip — rough soles, a dry surface, bodies low and leaning back — can pull harder before their feet slip. A team on a slippery surface will be dragged across, no matter how strong they are.

Coaches explain this with force arrows. They draw the rope, the two pulls and the friction on the feet, each as an arrow whose length shows its size. One look at the arrows shows which team is going to win.

C1. While the rope is not moving, what can you say about the two teams' pulls? Use the word *balanced*.

C2. What happens to the rope when one team pulls harder? Explain why, using the words *net force*.

C3. According to the passage, why does a team on a slippery surface lose, however strong they are?

C4. List two things from the passage that give a team better grip.

C5. How can a coach tell which team will win just by looking at the force arrows?

Part B • Theory and examples

2.1 Naming the Forces

Before you can draw a force diagram, you need the right names. These are the forces you will use again and again in this unit.

Force	What causes it	Which way it acts
Weight (W)	Gravity pulling on mass	Always straight down
Normal force (N)	A surface pushing back on an object	Perpendicular to the surface
Friction (f)	Two surfaces rubbing together	Opposite to the motion
Air resistance	Air pushing against a moving object	Opposite to the motion
Tension (T)	A rope, string or cable pulling	Along the rope, away from the object
Applied force	A person or machine pushing/pulling	Whichever way it is applied

2.2 Drawing Force Arrows — The Three Rules

The three rules for every force arrow

1. **Start the arrow on the object.** The tail sits on the object the force acts on.
2. **Point the arrowhead the way the force acts.** Down for weight, up for the normal force, and so on.
3. **Make the length match the size.** A 20 N force is drawn twice as long as a 10 N force. *Always.*

Rule 3 is the one students most often forget — and it matters, because the picture should tell you the answer at a glance. If two arrows are the same length, the forces are balanced. If one is longer, that is the direction the object accelerates.

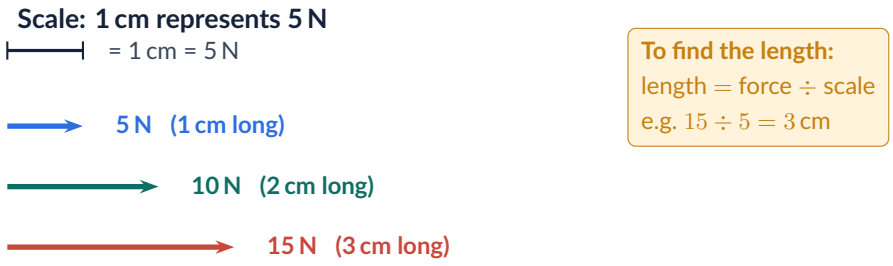


Figure 2.1 — Using a scale for force arrows. Here 1 cm represents 5 N, so a 15 N force is drawn 3 cm long.

Worked Example 2.1 — Drawing arrows to scale

Using a scale of 1 cm = 5 N, how long should the arrow be for a 25 N push?

Length = force ÷ scale = $25 \div 5 =$ **5** cm

2.3 Free Body Diagrams

A **free body diagram (FBD)** shows *all* the forces acting on *one* object, each drawn as a labelled arrow using the three rules. The object is drawn as a simple box or dot — nothing else from the picture is included.

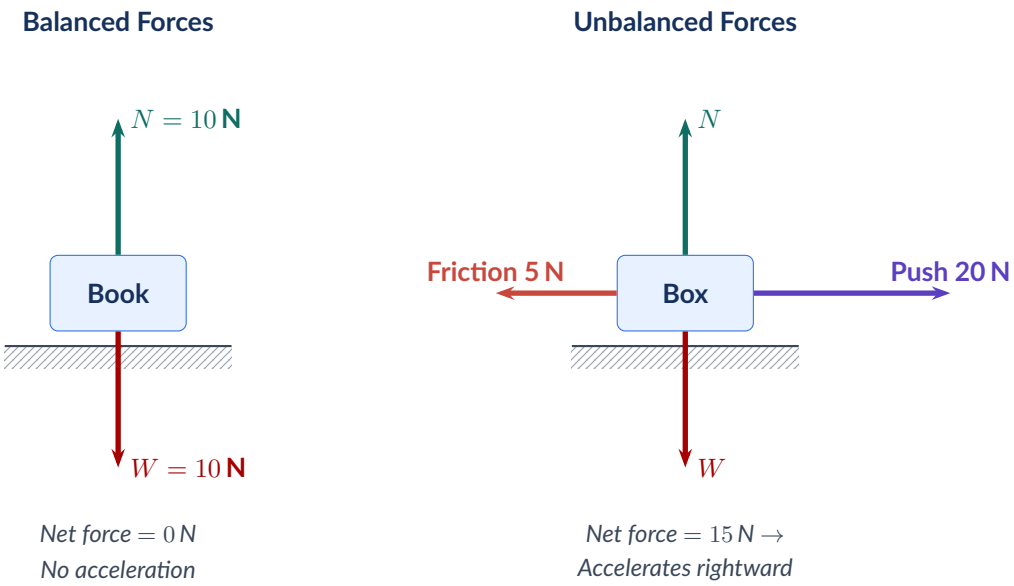


Figure 2.2 — Balanced forces (left) and unbalanced forces (right)

Worked Example 2.2 — Drawing a free body diagram

A book with a weight of 10 N rests on a table. Draw its free body diagram.

Step 1 — find every force. Gravity pulls the book down (*weight*). The table pushes up on it (*normal force*).
Step 2 — size each force. The book is not moving up or down, so the table pushes up exactly as hard as gravity pulls down: both are 10 N.
Step 3 — draw. A box for the book, a 10 N arrow down labelled W, and an arrow of the *same length* up labelled N — as in the left of Figure 2.2.

2.4 Balanced and Unbalanced: Net Force

When you add up all the forces on an object, the single force left over is called the **net force** (or resultant force).

Working out the net force

Same direction → **add** the forces.
Opposite directions → **subtract** the smaller from the larger. The net force acts in the direction of the larger force.

If the net force is **zero**, the forces are **balanced** and the motion does not change.
If the net force is **not zero**, the forces are **unbalanced** and the object speeds up, slows down or changes direction.

Worked Example 2.3 — Finding the net force

A sledge is pulled forwards with 45 N. Friction pushes back with 18 N. Find the net force.

Opposite directions, so subtract: $45 - 18 = \mathbf{27}$ N forwards

The net force is not zero, so the forces are unbalanced and the sledge **speeds up**.

Part C • Practice

Work through every task. Show your working for calculations, and always include the unit.

Task 2.1*(3 marks)* Using a scale of **1 cm = 2 N**, how long should the arrows be for forces of **(a)** 6 N, **(b)** 10 N and **(c)** 3 N?

Task 2.2*(4 marks)* A box is pulled right with 30 N and friction resists with 10 N. Using a scale of **1 cm = 10 N**, draw both arrows on the box below. Label each with its name and size.

Task 2.3a *(2 marks)* Look at the balanced forces diagram in Figure 2.2 (left). Why is the book not speeding up or slowing down? Use the words *net force* in your answer.

Task 2.3b *(2 marks)* Look at the unbalanced forces diagram (right). Calculate the net force on the box and state which way it speeds up.

Task 2.3c *(2 marks)* Draw a free body diagram for a skydiver falling at a steady speed (terminal velocity). Label each force with its name and show their comparative sizes.

Draw your force diagram here.

Task 2.4 *(2 marks)* A lamp with a weight of 15 N hangs still from the ceiling on a cord. Draw its free body diagram, labelling each force with its name and size.

Draw your force diagram here.

Net force. For each question, state the size *and* direction of the net force, and whether the forces are balanced or unbalanced.

Task 2.5 *(2 marks)*
Two people push a car in the same direction with 250 N and 310 N. Find the net force.

Task 2.6 *(2 marks)*
A trolley is pushed right with 40 N while friction pushes left with 12 N. Find the net force.

Task 2.7 *(2 marks)*
A book rests on a table. Its weight is 8 N. What is the normal force from the table? Explain.

Task 2.8 *(2 marks)*
In a tug-of-war, one team pulls left with 900 N and the other pulls right with 900 N. Describe the motion of the rope.

Task 2.9 *(3 marks)* A rocket engine produces 50 000 N of upward thrust. The rocket's weight is 32 000 N. **(a)** Find the net force. **(b)** State which way the rocket speeds up. **(c)** Explain what would happen if the thrust dropped to 32 000 N.

Task 2.10 *(2 marks)* A swimmer moves forward at a steady speed. Her arms provide 120 N of forward force. **(a)** What is the size of the water resistance acting on her? **(b)** Explain how you know.

Task 2.11 *(3 marks)* A parachutist has a weight of 700 N. Just after the parachute opens, air resistance is 950 N. **(a)** Find the net force and its direction. **(b)** Describe what happens to the parachutist's speed. **(c)** Air resistance then falls to 700 N — describe the motion now.

Part D • Above level

These questions go beyond the Year 7 standard. They ask you to combine ideas, use a formula, or judge someone else's reasoning.

Extension 2.12 *(3 marks)* Sam says: *"A hockey puck sliding across ice at a steady speed must have an unbalanced force pushing it forwards."* Is Sam right? Use a free body diagram and the idea of net force to explain.

Extension 2.13 *(4 marks)* A skydiver jumps from a plane. Draw a free body diagram for each moment: **(i)** just after jumping, **(ii)** while still speeding up, **(iii)** at terminal velocity. Then explain why air resistance grows, and why the skydiver eventually stops speeding up.

(i) just after jumping

(ii) speeding up

(iii) terminal velocity

Extension 2.14 *(3 marks)* A car has a driving force of 4500 N and a total resistive force of 1200 N. Calculate the net force, state which way the car accelerates, and suggest one way the driver could make the forces balanced.

Extension 2.15 *(4 marks)* A 1200 kg car has a driving force of 4800 N. Air resistance and friction total 1500 N. **(a)** Find the net force. **(b)** Use $F = ma$ (acceleration = net force ÷ mass) to find the acceleration. **(c)** The driving force rises to 6000 N — find the new acceleration.

3 Experiment 1: Investigating Friction

Prediction • Fair testing • Repeat trials • Means

Victorian Curriculum 2.0: **VC2S8U14 • VC2S8I01 • VC2S8I02 • VC2S8I03 • VC2S8I04 • VC2S8I05**

Part A • Reading and comprehension

Reading — Grip

Friction is the force that makes walking possible. With every step, your shoe pushes backwards against the ground, and friction pushes you forwards. On ice, where friction is tiny, your foot slips backwards and you go nowhere.

Friction comes from the roughness of surfaces. Even surfaces that look smooth are covered in tiny bumps when you look at them under a microscope. When two surfaces press together, the bumps catch on each other and resist sliding. Rougher surfaces have bigger bumps that catch more, so they produce more friction. Pressing the surfaces together harder also increases friction.

Sometimes engineers want as much friction as possible — in the rubber of a car tyre, the soles of running shoes or the grip tape on a skateboard. Other times they want as little as possible: oil in an engine and wax on skis reduce friction, so parts slide easily and less energy is wasted as heat.

In this experiment you will measure friction with a spring scale. When you pull a block along at a *steady* speed, your pull and the friction are balanced — so the reading on the spring scale is equal to the friction force.

C1. Explain how friction helps you to walk, and why walking on ice is so hard.

C2. Why do rougher surfaces produce more friction? Use the word *bumps* in your answer.

C3. Give one example where engineers want more friction, and one where they want less.

C4. Why must the block move at a *steady* speed when you read the spring scale?

Part B • Planning the experiment

Experiment 1: Investigating Friction

Curriculum: VC2S8U14 • VC2S8I01 • VC2S8I02 • VC2S8I03 • VC2S8I04 • VC2S8I05
Aim: *To investigate how the type of surface affects the friction force on a sliding object.*

3.1 Making it a Fair Test

In a fair test you change **one** thing, measure **one** thing, and keep everything else the same.

Variable	What it means
Independent	The one thing *you* change on purpose
Dependent	The thing you *measure* to see the effect
Controlled	Everything you keep the *same*, so the test is fair

Worked Example 3.1 — Identifying variables

A student rolls a toy car down a ramp at different heights and measures how far it travels across the floor. Identify the variables.

Independent: the height of the ramp (the student changes it).
Dependent: the distance the car travels (the student measures it).
Controlled: the same car, the same floor surface, releasing the car without pushing it.

Prediction — before you begin

Which surface do you think will produce the most friction? Why?

Equipment

Wooden block (or heavy book) Spring scale (newton meter) 3+ surfaces (bench, sandpaper, fabric, carpet) Ruler

Method

1. Place the wooden block on the first surface.
2. Attach the spring scale to the block. Pull slowly and steadily.
3. Read the spring scale while the block moves at a steady speed. Record the friction force in newtons.
4. Repeat 3 times for this surface and record all values.
5. Repeat steps 1–4 for each surface.

Safety

Pull gently and keep the scale in line with the block — a spring scale that is overstretched can snap back. Keep sandpaper away from eyes and skin.

Part C • Results and analysis

Results

Surface	Trial 1 (N)	Trial 2 (N)	Trial 3 (N)	Mean (N)

Calculating a mean

Add the trials, then divide by how many there are. For 1.2 N, 1.4 N and 1.3 N: $(1.2 + 1.4 + 1.3) \div 3 = 3.9 \div 3 =$ **1.3** N

Task 3.1 *(3 marks)* Identify the independent variable, the dependent variable, and two controlled variables in *your* experiment.

Task 3.2 *(2 marks)* Which surface produced the greatest friction? Which produced the least? Was your prediction correct?

Task 3.3 *(2 marks)* Why did you take three trials for each surface? What did you do with the three values?

Task 3.4 *(2 marks)*
Another group measured 2.1 N, 2.3 N and 2.2 N on carpet. Calculate the mean.

Task 3.5 *(2 marks)*
A group measured 1.8 N, 1.9 N and 2.9 N on fabric. Which value looks wrong? Find the mean of the other two.

Task 3.6 *(2 marks)* Use the reading on page 16 to explain *why* the surface you named in Task 3.2 produced the most friction.

Part D • Above level

These questions go beyond the Year 7 standard. They ask you to evaluate the method, predict, or use a formula.

Extension 3.7 *(2 marks)* Even on the same surface, your three readings were probably not identical. Suggest one reason why, and one change to the method that would make the readings more consistent.

Extension 3.8 *(2 marks)* Predict what would happen to the friction force if you stacked a second block on top of the first. Explain your prediction using the reading.

Extension 3.9 *(3 marks)* The wooden block has a mass of 0.3 kg. Choose your highest mean friction force. If the block were sliding and you let go of the spring scale, friction would be the only horizontal force on it. Use $F = ma$ (acceleration = force ÷ mass) to calculate how quickly it would slow down. Show all working.